Star Tracker PRO

Technical Operations & Telemetry Manual

Ai One

What This Is

Star Tracker PRO is a real-time sky map and deep-space telemetry terminal, driven by your actual location and the current time — not stock imagery or a static almanac. Every position on screen is computed live via client-side astronomical algorithms (astronomy-engine for the Sun/Moon/planets, a local SGP4 propagator for the ISS), paired with live data feeds from NASA/JPL and NOAA.

Getting Oriented

- The default view opens at 1.0x zoom / 180° field of view — the full sky dome, centered on the Zenith (straight overhead), so you keep full contextual awareness of the sky rather than starting zoomed into one target.
- The outer edge of the dome is your horizon. N / E / S / W around it mark compass direction along that horizon. Azimuth increases clockwise from North (0°) through East (90°), South (180°), and West (270°).
- Drag anywhere on the dome to pan. Scroll or pinch to zoom (1x–5x). Double-click (or double-tap) the dome to reset pan and zoom back to the default wide view.
- Tapping a Sun/Moon/planet marker selects and highlights it without changing your zoom level — the dome stays wide unless you zoom in yourself.

Observatory Horizon Nodes

By default, Star Tracker PRO uses your live location (GPS, or an approximate IP-based location if GPS isn't available/granted). You can instead view the sky from a real preset observatory via the observatory picker near the top of the view:

- Mauna Kea Observatories — Hawaiʻi, USA
- Palomar Observatory — California, USA
- Royal Observatory Greenwich — London, UK
- Griffith Observatory — Los Angeles, USA
- Very Large Telescope (VLT) — Atacama, Chile

Each preset uses that site's real coordinates and elevation for horizon/topocentric recalculation.

Tracking the ISS

Toggle the "ISS" layer to plot the International Space Station's live position on the dome, including a telemetry readout (elevation, azimuth, altitude, velocity, range). The same toggle also enables an embedded live NASA video feed from the ISS. This uses real orbital propagation, not a scripted path.

Celestial Bodies & the Casual/Expert Toggle

Tap any Sun/Moon/planet marker to see its details. Use the Casual/Expert switch to control how much you see:

- **Casual** — a plain-language summary: what it is, which direction to look, and roughly how far away it is.
- **Expert** — the full technical readout: azimuth, altitude, magnitude, precise distance, and next rise/set times.

Both modes show the exact same real, live-computed data — Casual just shows less of it at once.

Sky Maps Overlay

The Sky Maps selector overlays real constellation line data: Big Dipper/Polaris, the Zodiac/ecliptic, the brightest stars, Messier deep-sky objects (nebulae, galaxies, star clusters), or all constellations at once.

Live NOAA Space Weather

Toggle Space Weather to see a live NOAA geomagnetic Kp index reading with a human-readable storm-level status — real data from NOAA's public feed, refreshed periodically.

Sky Fest

Sky Fest opens a panel of three tabs above the event feed:

DSN Telemetry

Live status from NASA/JPL's real Deep Space Network: which ground antennas at Goldstone, Madrid, and Canberra are currently locked onto which spacecraft, each link's band and data rate, signal power, and one-way light-time (computed from the network's own real distance figures for that spacecraft). A spacecraft not currently shown simply isn't in active contact with any antenna right now.

Deep Sky Spectrum

Real multi-wavelength survey imagery — Optical, Infrared, Radio, and X-Ray — for whichever Messier object you have selected on the sky map, sourced live from NASA's SkyView Virtual Observatory. Alongside the imagery, Kali provides a short AI-generated summary of the object; this is explicitly commentary, not a precise catalog measurement.

Space Media

Paste a YouTube link to save and play space-related video (rover footage, telescope livestreams, mission highlights) inline.

HUD Controls

Once a telescope is connected (see below), a HUD Controls panel offers:

- **Tracking Rate** — Sidereal / Solar / Lunar / Stopped presets. These send real mount commands where your connected protocol supports them; presets a given protocol has no real command for are shown disabled rather than silently doing nothing.
- **Sensor Gain** — shown disabled. This build has no camera/sensor hardware integration, so there is no real gain value to report or control.
- **HUD Opacity** — a local display preference for the telemetry overlay's own transparency.

Telescope Connection (Optional)

If you own a compatible telescope mount, Star Tracker PRO can connect to it directly over Web Serial or Web Bluetooth (Chrome/Edge), using real LX200/NexStar protocol support — live position polling, goto/slew commands, and tracking-rate control. A Simulator mode is also available if you want to try the interface without real hardware connected.

Questions

Something not working as expected? Reach out through the Ai One Support drawer in the Vault, or via the site's contact channels.